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Laminated mica nanosheets supported ionic liquid membrane for CO_2 separation

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Abstract

Mica has attracted great attention due to its excellent photoelectric property and it is mainly used in the optics and electronics area. High-quality mica nanosheets exfoliated from natural ground mica have been successfully achieved recently. It improves the application potential of mica as well as extending the application field to gas separation as a new two-dimensional (2D) material. Herein, we firstly constructed a mica membrane by regularly stacking mica nanosheets and then immobilized ionic liquid (IL) into its 2D channels to separate CO_2 from H_2, CH_4 and N_2. Gas diffusion is changed from Knudsen diffusion in the mica membrane to a solution–diffusion mechanism after inserting IL. The resultant membrane, the mica supported IL membrane (M-SILM), has about 80 GPU for CO_2 permeance, and selectivity for CO_2/H_2, CO_2/CH_4 and CO_2/N_2 of 7.7, 28.6 and 87, respectively. It is the first time a mica nanosheet to construct a gas separation membrane has been used. As a cheap raw material with facile treating, mica shows promising potential for gas separation and competitiveness among 2D material supported IL membranes.

Supplementary material for this article is available online

Keywords: CO_2 separation, nanoconfined ionic liquid, mica supported ionic liquid membrane

(Some figures may appear in colour only in the online journal)

1. Introduction

One of the greenhouse gases, CO_2, whose emission from the combustion of fuels, has greatly aggravated global warming and other climate anomalies [1]. Efficient CO_2 capture and storage meets the consensus of all mankind, sustainable development. The three conventional methods, amine-based absorption, porous solids-based adsorption and cryogenic distillation have taken up most of the mission. However, their high energy consumption or high cost causes a disregard [2–4]. So, a membrane separation technique, with low energy consumption and high efficiency [5], has received more attention. But there is still a long way to put into practice because of its preparation cost and trade-off between permeance and selectivity [6].

Various promising new materials such as metal organic frameworks (MOFs) [7, 8], two-dimensional (2D) material [9, 10] and ionic liquid (IL) [11, 12] are developed to overcome the above drawbacks. Each of them has unique property, including a designable pore size for MOFs, ultrathin membrane for 2D material and specific adsorption of CO_2 for IL. In addition, IL is considered an attractive alternative for CO_2 capture and storage because of its extremely low volatility, pretty high thermal stability and easily tunable physicochemical property [13, 14]. However, no matter how excellent the property is, the separation membrane prepared by a single material can only have limited improvement in performance, so composite membranes are the megatrends.

[3] These authors contributed equally to this work.

The limited usage of IL because of its liquid phase also urges the preparation of IL-based composite membranes.

Using a thin layer IL to cover the pore surface of a porous medium or confining IL into a porous substrate are two common strategies, forming a supported IL phase (SILP) [15] and supported IL membrane (SILM) [16], respectively. However, leakage of the IL is unavoidable under high pressure. A thicker membrane or smaller pore size can alleviate this. While a thicker membrane always brings about smaller permeance, a smaller pore size may be better. Much progress has been achieved, such as an IL functional graphene membrane [17] and MOF confined IL membrane both have shown enhanced CO_2 separation [18]. In addition, when pore size comes to the nanoscale, the unusual properties of IL come out [19], such as the anomalous thermal property [20], increase of CO_2 adsorption or diffusion [21, 22] and spatial distribution of ions [23, 24]. However, most work focus on the one-dimensional nanotube or three-dimensional porous network, only a little has research on IL confined in two-dimensional (2D) nanochannels, such as graphene oxide (GO) [25], and WS_2 laminated membranes [26], has been done. Although the above membranes show excellent CO_2 separation performance, the preparation process of GO or WS_2 nanosheets is relatively complex and costly.

Recently, ground mica has been exfoliated into high-quality mono-layered or few-layered mica nanosheets [27] and widely used in electronics because of its high dielectric strength, ultraviolet-shielding and high chemical and thermal stability [28, 29]. However, to the best of our knowledge, it is an unworked field for mica to separate gases, except for gas adsorption [30]. Due to the cheap raw material and facile exfoliating process, mica nanosheets provide a new possibility for constructing novel laminated gas separation membranes.

Herein, we stacked mica nanosheets and confined IL, 1-butyl-3-methylimidazolium tetrafluoroborate ([BMIM][BF_4]), into the 2D nanochannels between the mica nanosheets to form a mica supported IL membrane (M-SILM). The resultant M-SILM shows a permeance of CO_2 about 80 gas permeation units (GPU) with high selectivity for CO_2 over other light gases such as H_2, CH_4 and N_2. The low-cost and facile preparation process indicates M-SILM holds great potential for gas separation.

2. Method

2.1. Synthesis of mica membrane and M-SILM

Mica nanosheets were exfoliated by intercalating cetyltrimethylammonium bromide (CTAB) into pretreatment ground mica, as reported elsewhere [27]. Specifically, the original mica powder (S0) was pretreated by heating at 800 °C for 1 h. Then 3 g pretreated mica power (S1) was added into 100 ml 5 mol l^{-1} nitric acid and stirred at 95 °C in a thermostatic water bath for 5 h. The resultant product was washed by deionized water to neutral through a filter and dried at 80 °C. Then the dried powder (S2) was reacted with a NaCl supersaturated solution at 95 °C for 3 h, washed to remove the rest of the NaCl and filtrated again, and dried at 80 °C (S3). Subsequently, product S3 and a certain amount of surfactant CTAB were dispersed in deionized water with a mass ratio of 3:100, and stirred at 80 °C for 24 h to intercalate the CTAB into the interlayer of mica. Finally, after washing, filtering and drying, the intercalated mica product (S4) was further exfoliated by ultrasonication treatment in ethanol. 0.1 g S4 was added into 100 ml ethanol, and ultrasonicating for 4 h under a sonication power of 600 W. After being centrifuged for 10 min at the speed of 6000 r m^{-1} twice, the supernatant was a mica nanosheets ethanol solution.

The mica membrane was prepared by vacuum filtrating the above solution on an anodic alumina oxide (AAO). M-SILM was prepared by drop-casting a certain volume of IL on the surface of the mica membrane. After 2 h, the residue IL was span away under 7000 r m^{-1} for 60 s.

2.2. Gas separation

Gas permeance of the mica membrane and M-SILM was tested by a bubble flow meter with total volume of 1 ml and 0.01 ml accuracy. The membrane was immobilized in an in-line stainless steel filter holder, purchased from Millipore, with effective membrane surface area 2.2 cm^2. Each data point was measured five times and the average value taken. Gas permeance (Q) (with standard unit $L \cdot m^{-2} s^{-1} Pa^{-1}$) was calculated by equation (1) [25]:

$$Q = \frac{1}{\Delta P} * \frac{273.15}{273.15 + T} * \frac{P_{atm}}{101.325\ \mathrm{kPa}} * \frac{1}{A} * \frac{dV}{dt}, \quad (1)$$

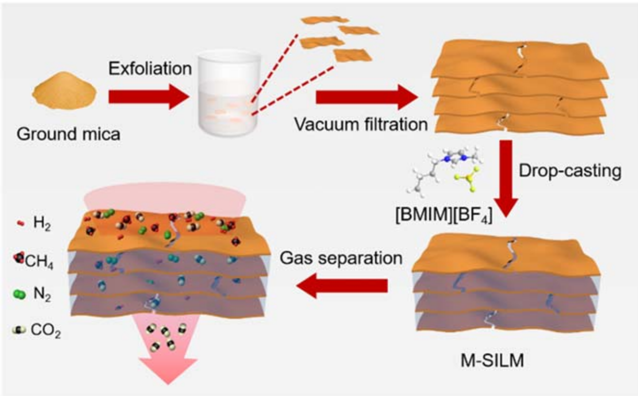
where ΔP, P_{atm}, A, T, dV/dt are the pressure difference on the membrane, atmospheric pressure, membrane effective area, temperature and volumetric displacement rate in the bubble flow meter, respectively. The ideal selectivity of two gases is the ratio of two gases permeance, as calculated by equation (2) [25]:

$$\alpha = \frac{Q_1}{Q_2} \quad (2)$$

where Q_1 and Q_2 are the two gases permeance, respectively. The gas permeability was calculated by multiplying permeance (Q) and thickness.

2.3. Characterization

Scanning electron microscope (SEM) images were obtained by a Hitachi S-4800 field-emission SEM equipped with an energy dispersive x-ray spectrometer. Atomic force microscopy (AFM) images were obtained by a Bruker Dimension Edge and the Fourier-transform infrared (FTIR) spectrum was obtained by a Bruker Tensor 27. X-ray diffraction (XRD) patterns were obtained by a SHIMADZU XRD-7000 with a Cu Kα radiation source. Differential scanning calorimeter (DSC) results were obtained by a TA Q200. Transmission electron microscopy (TEM) images were obtained by a HT-7700.



Scheme 1. The preparation process and gas separation of M-SILM.

3. Results and discussion

3.1. Characterization of M-SILM

The scheme of preparation and test of M-SILM is shown in scheme 1. Compared with other 2D materials, such as GO, the price of the original material, ground mica, is less than half of graphite. The other materials, NaCl, CTAB or dilute nitric acid are also very cheap and safer than the material used in the GO preparation process. XRD patterns of the products under different stages of the exfoliation process are shown in figure S1, available online at stacks.iop.org/NANO/30/385705/mmedia. The intensity of reflections peaks weakened and became wider from S0 to S4 during the exfoliation process (The details can be seen in the section 2), and the reflections corresponding to 2θ moved to a lower degree, indicating the distance between the mica layer along (002) is increased, which is consistent with the reported literature [31]. It shows the evolution of exfoliation. The AFM image in figure S2 shows the size of a single-layer mica nanosheet is several hundred nanometers, with a height of about 1 nm. The morphology is shown in the TEM image in figure S3 and SEM images in figures 1(a) and (c). Regular stacks of mica nanosheets led to the apparent layered structure on the cross-section, and it is dense and orderly. [BMIM][BF_4] ionic liquid was confined in the mica membrane through capillary force. After incorporation of [BMIM][BF_4], the surface of the mica membrane became smoother and the layered structure cannot be observed anymore (figures 1(b) and (d)). As [BMIM][BF_4] is a liquid and fulfills all the channels of laminated mica nanosheets, the layered structure on the cross-section is immersed by the IL. While the strain induced wrinkles on the surface would be released due to the incorporated IL. XRD patterns of the mica membrane and M-SILM are displayed in figure 2(a). The peak corresponding to d_{002} of M-SILM widens and has obvious shifts to a lower degree compared with the mica membrane, as a result of IL insertion. The membrane thickness thus is expanded from 980 to 1140 nm. The uniform distribution of B, N and F in the mapping images of M-SILM (figure S4) shows the uniform incorporation of [BMIM][BF_4] in the M-SILM. The peak shifts between pure

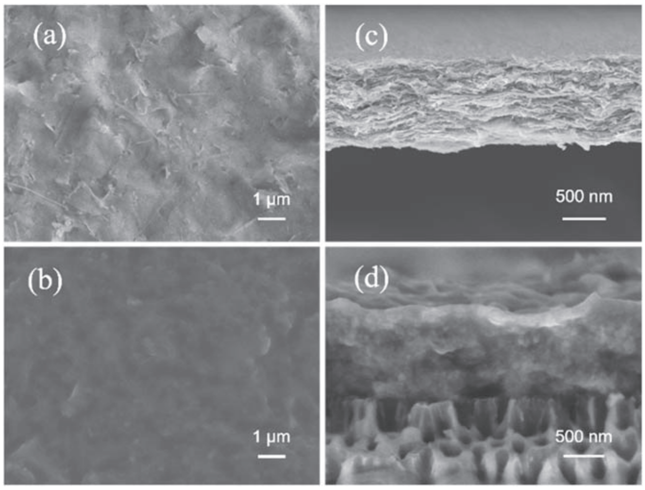


Figure 1. SEM images of the mica membrane and M-SILM and mapping image of M-SILM. (a) and (c) Surface of the mica membrane and M-SILM, respectively; (b) and (d) Cross-section of the mica membrane and M-SILM, respectively.

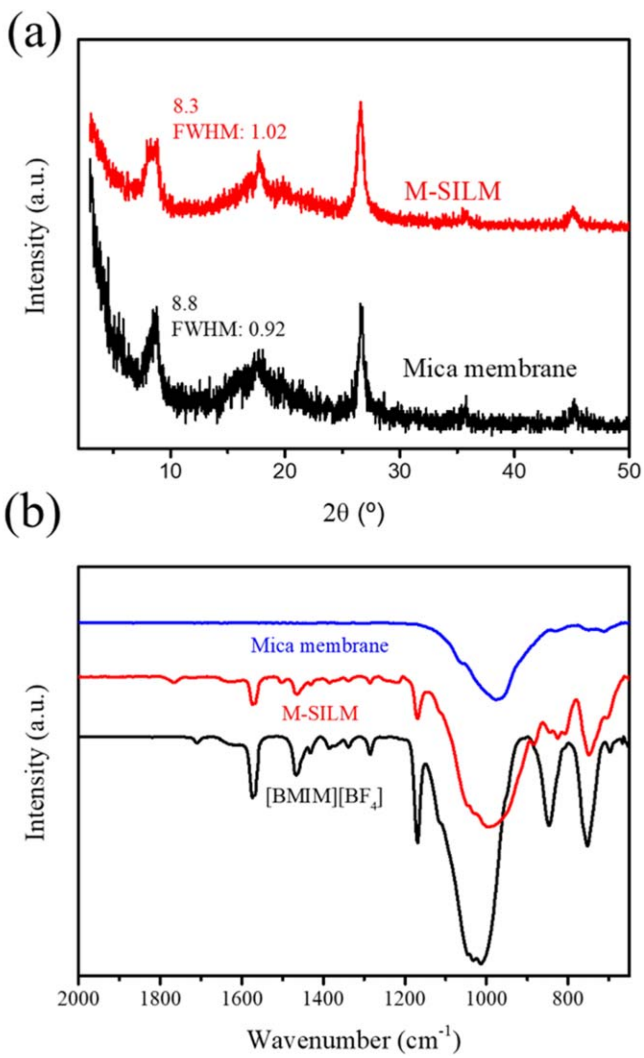


Figure 2. (a) The XRD patterns of the mica membrane and M-SILM; (b) the FTIR spectra of the mica membrane, [BMIM][BF_4] and M-SILM, respectively.

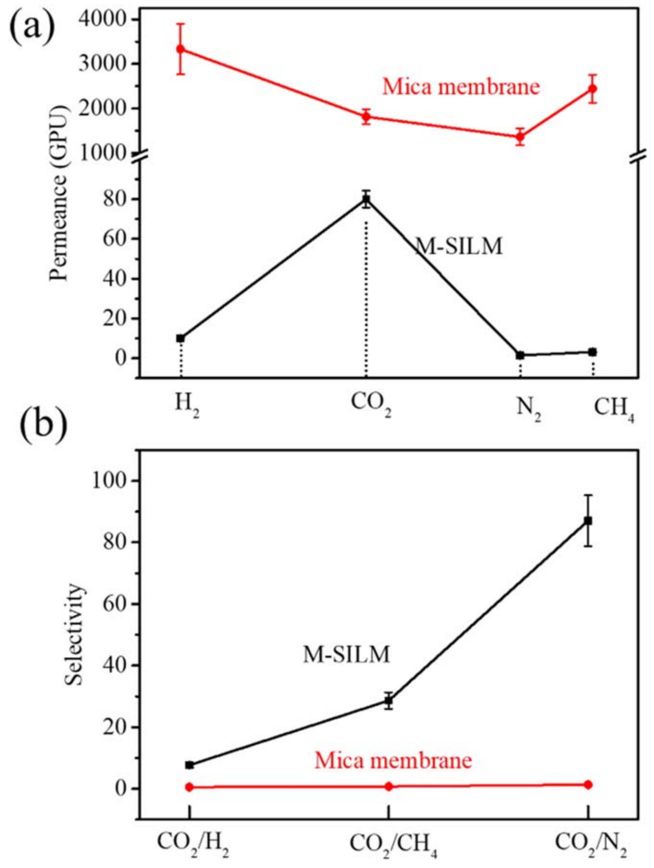


Figure 3. The gas separation performance of the mica membrane and M-SILM. (a) Four gases permeance and (b) gas selectivity.

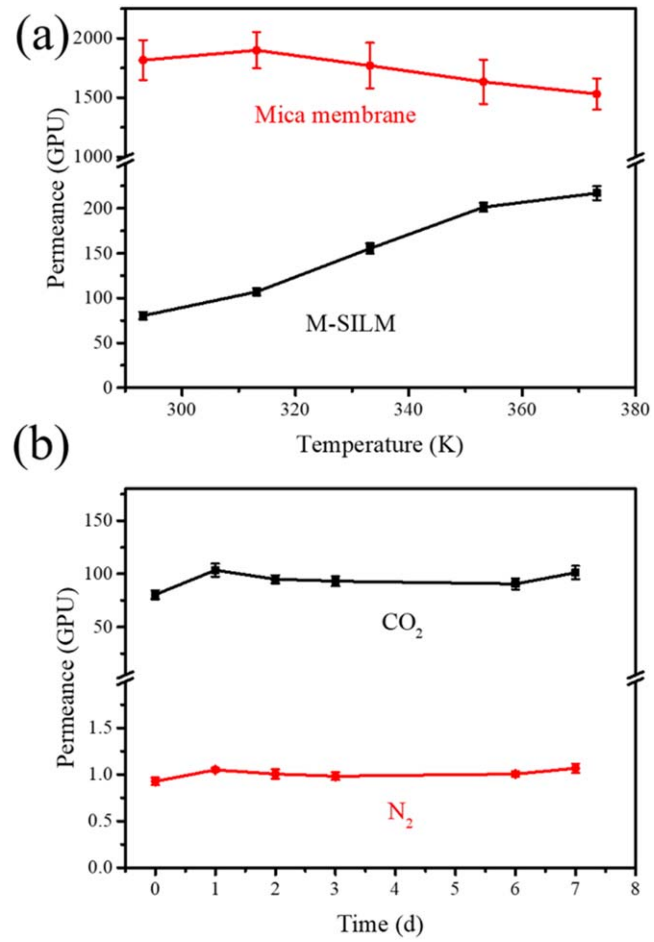


Figure 4. The stability of M-SILM. (a) CO_2 permeance of the mica membrane and M-SILM under 293.15 K$\sim$373.15 K; (b) CO_2 and N_2 permeance of M-SILM tested for a week.

IL and IL in the M-SILM on the FTIR spectra (figure 2(b) and table S1) reflects the interaction between mica nanosheets and IL and confirms that IL is confined in the M-SILM. The major shifts come from these peaks corresponding to anion of IL, BF_4^-, and N in the imidazole ring. It reflects the interaction among mica nanosheets, anions or imidazole ring of IL or the influence of nanoconfined IL on its anions or imidazole ring. Figure S5 shows the DSC curves of mica membrane, [BMIM][BF_4] and M-SILM, and the freezing point of IL decreases from -73 °C to -77 °C. It also confirms the nanoconfinement of IL as mentioned above that nanoconfined IL may have anomalous thermal property.

3.2. Gas separation

The nanometer scale channels stacked by mica nanosheets forms the path for gas transport, and the mechanism of gas through the mica membrane conforms to the Knudsen diffusion model [32], which is relevant to the gas mean free path and membrane pore size.

The resultant membrane was fixed in an in-line stainless steel filter holder and gas permeance was tested by a bubble flow meter. As figure 3 shows, the mica membrane with a thickness of 980 nm has the largest permeance for H_2 as its smallest gas molecular weight and dynamic diameter, while secondly for CH_4 and the smallest for N_2. The selectivity of CO_2/H_2, CO_2/CH_4 and CO_2/N_2 is 0.5, 0.74 and 1.33, respectively. The deviation from Knudsen selectivity may be a result of the defects in the membrane. However, when IL is nanoconfined into mica nanochannels, CO_2 separation performance shows a big lift, the selectivity is 7.7, 28.6 and 87 for CO_2/H_2, CO_2/CH_4 and CO_2/N_2, respectively. All four gases permeance have a large drop compared with pure mica membrane, but CO_2 permeance shows the smallest variation. This is because IL fully fills M-SILM's nanochannels and gases can only transport through IL. Hence, the mechanism of gas transport changes from Knudsen diffusion to solution–diffusion [33]. The gas solubility and diffusivity of IL determine the permeance. Previous work has confirmed that [BMIM][BF_4] has the largest solubility for CO_2 among H_2, CH_4, N_2 and CO_2 [34], and the nanoconfinement of [BMIM][BF_4] further increases CO_2 solubility while least decreases CO_2 diffusivity [25]. Specifically, the negative-charged mica [27] nanosheets will attract cations of IL and result in the separation of anions and cations. It is well known that CO_2 has a better affinity with anions [25, 35], so the relaxed anions play the role of a carrier to form a fast transmission network for CO_2. Therefore, M-SILM shows the above excellent CO_2 separation performance, which is comparable to those reported SILMs and close to the 2018 Robeson upper bound [6] (figure S6 and table S2).

When the membrane thickness decreases to 520 nm, gases permeance is increased relatively, while the selectivities is declined somewhat. Especially, when membrane thickness comes to 210 nm, gas permeance obviously increases, but the selectivity decreases significantly (figure S7). In addition, if the membrane is too thin, the durability of the membrane will also be another issue. Therefore, we looked at the 1140 nm thick M-SILM in a detailed investigation.

We also tested the temperature and time durability of M-SILM. Figure 4(a) shows CO_2 permeance of the pure mica membrane and M-SILM under 293.15 K~373.15 K. CO_2 permeance of the mica membrane decreases with increasing temperature, while CO_2 permeance of M-SILM increases. This is because the increase of CO_2 diffusivity has more influence than the decrease of CO_2 solubility [25]. However, the selectivity of CO_2/N_2 decreases to 70% when the temperature reaches 373.15 K. As for long-time durability, we tested CO_2 and N_2 permeance of M-SILM for a week. The permeance has no obvious change. It confirms the good stability of M-SILM (figure 4(b)).

Except AAO as a substrate, we also tested the M-SILM membrane assembled on the polycarbonate porous membrane with a pore size of 200 nm. Almost the same performance is achieved as that assembled on AAO (figure S8). This means the pore substrate only works as a support for assembling M-SILM, without an effect on the gas separation performance.

4. Conclusion

In summary, we use mica nanosheets, exfoliated by CTAB, to construct a mica nanosheets laminated membrane for gas separation by filling IL into the 2D nanochannels of the laminated mica membrane. The negative-charged mica nanosheets are attracted to the cations of IL, which relaxed the anions to form the network for CO_2 transportation. Hence, the resultant M-SILM presents good CO_2 separation performance based on the solution–diffusion mechanism.

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